

1. Setup and Utilities

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Template

```
1 #include "bits/stdc++.h"
2 using namespace std;
3 using ll = long long;
4 #define int ll
5 #define endl '\n'
6 void solve() {
7 }
8 signed main() {
9     cin.tie(0)->sync_with_stdio(0);
10    cin.exceptions(cin.failbit);
11    int tt = 1;
12    // cin >> tt;
13    while (tt--) {
14        solve();
15        cout << '\n';
16    }
17    return 0;
18 }
```

Precompile stdc++.h

```
1 sudo g++ -x c++-header -std=c++17 -O0 stdc++.h -o stdc++.h.gch
2 ulimit -s ${size in kb}
3 add empty template to live templates
4 increase undo range to 10k (ctrl+shift+A) -> registry -> undo
```

Pragmas

```
1 #pragma GCC optimize("Ofast")
2 #pragma GCC optimize ("unroll-loops")
3 #pragma GCC target("sse,sse2,sse3,ssse3,sse4,popcnt,abm,mmx,avx,tune=native")
```

Random mt19937

```
1 mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2 ll rnd(ll l, ll r) {
3     static mt19937_64 gen(chrono::steady_clock::now().time_since_epoch().count());
4     return uniform_int_distribution<ll>(l, r)(gen);
5 }
```

Direction Arrays

```
1 int dx[8] = { 2, 1, -1, -2, -2, -1, 1, 2 };
2 int dy[8] = { 1, 2, 2, 1, -1, -2, -2, -1 }; // knight
3
4 int dx[8] = {-1,0,1,-1,1,-1,0,1};
5 int dy[8] = {-1,-1,-1,0,0,1,1,1}; // king
6
7 int dx[4] = {1, -1, 0, 0};
8 int dy[4] = {0, 0, -1, 1};
9 string direction = "DULR";
```

Custom Comparator

```
1 struct cmp {
2     bool operator() (int a, int b) const {
3         return ...;
4     }
5 };
6 set<int, cmp> s;
```

Fast HashMap (gp_hash_table)

```
1 #include <ext/pb_ds/assoc_container.hpp>
2 using namespace __gnu_pbds;
3 struct chash {
4     const int RANDOM = (ll)(make_unique<char>().get()) ^
5         chrono::high_resolution_clock::now().time_since_epoch().count();
6
7     static unsigned long long hash_f(unsigned long long x) {
8         x += 0x9e3779b97f4a7c15;
9         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
10        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
11        return x ^ (x >> 31);
12    }
13    static unsigned hash_combine(unsigned a, unsigned b) {
14        return a * 31 + b;
```

```

15     }
16     int operator()(int x) const {
17         return hash_f(x)^RANDOM;
18     }
19 };
20 gp_hash_table<int, int, chash> table;

```

__int128_t

```

1  istream& operator>>(istream& is, __int128_t& v) {
2      string s; is >> s; v = 0;
3      for (char c : s) if (isdigit(c)) v = v * 10 + c - '0';
4      if (s[0] == '-') v = -v;
5      return is;
6  }
7  ostream& operator<<(ostream& os, __int128_t v) {
8      if (!v) return os << "0";
9      if (v < 0) os << '-', v = -v;
10     string s;
11     while (v) s += '0' + v % 10, v /= 10;
12     reverse(s.begin(), s.end());
13     return os << s;
14 }

```